REVERSE ENGINEERING

Dell Wired USB Optical Mouse

Reverse-engineer the device as a compact USB HID system for 2D cursor movement and scrolling.



LEFT / RIGHT BUTTONS: Mechanical click input

SCROLL WHEEL: Scrolling + wheel button

OPTICAL SENSOR: Tracks surface movement

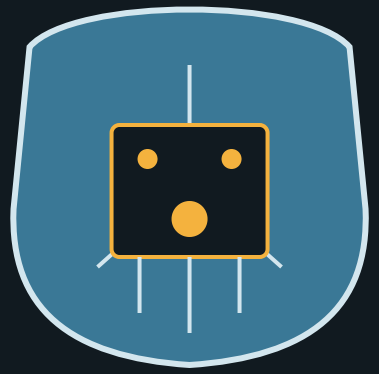
PURPOSE: 2D cursor movement + scrolling

INTERFACE: USB HID

POWER: 5V DC through USB

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SPECIFICATION DASHBOARD



Technical Specifications

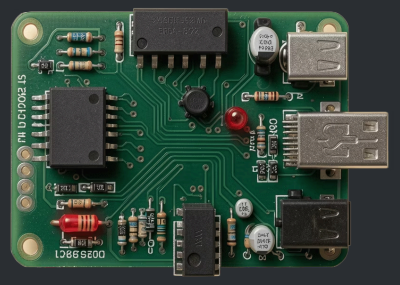
POWER 5V DC	INTERFACE USB HID	USB CONNECTION VCC | GND | D+ | D-
PCB Single-sided FR-2	PCB THICKNESS 1.6 mm	OPTICAL IC 8-pin optical DIP
PROCESSING Integrated DSP + USB PHY	SWITCH RATING 5V DC, 50mA	SWITCH LIFE 1,000,000+ cycles

Present as a specification dashboard with a simplified mouse/PCB illustration and compact icon-led cards.

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BOARD VIEW

Internal Components & Their Functions



812-00954-01A / SM8122_UB

1 – **U1:** Optical Sensor & Controller IC

2 – **D1:** Red/IR Illumination LED

3 – **SW1 / SW2:** Left & Right Tactile Switches

4 – **BM1:** Scroll Wheel Button

5 – **ENC1:** Rotary Scroll Encoder

6 – **R1–R8:** Carbon Film Resistors

7 – **C4 / C5:** Power Decoupling Capacitors

8 – **J1:** 4-Pin USB Connector

LEGEND

INPUT → PROCESSING → POWER FILTERING → USB OUTPUT

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ENGINEERING PROCESS

How the Optical Mouse Works

01

USB POWER

USB provides 5V DC

02

SURFACE ILLUMINATION

LED D1 illuminates the surface through the optical lens.

03

IMAGE CAPTURE

Optical sensor U1 captures microscopic surface patterns.

04

MOTION PROCESSING

Internal DSP compares consecutive images: Image A → Image B → ΔX / ΔY

05

INPUT DETECTION

Switches and encoder detect left/right clicks, scroll movement, and scroll-button press.

06

USB DATA

U1 packages the information into USB HID data and sends it through D+ / D-.

Surface Movement → Optical Sensor → DSP → USB → Computer Cursor

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SYSTEM ARCHITECTURE

Circuit Diagram

https://www.tinkercad.com/things/1b9VcWA35Fu/editel?returnTo=%2Fdashboard

Report ——

https://go.fliplink.me/view/BB66B240-747D-4A1D-ACAC-299F6E840BB1

A single optical IC integrates sensing, motion processing, input handling and USB communication.

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